Exercise Sheet

Lecture 2

Span, Linear Independence, Basis, Dimension

Throughout, $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Tags indicate the flavour of each problem; a $\star$ marks a harder one. Deciding membership in a span, testing independence, and finding coordinates all reduce to solving linear systems.

Core tutorial route

Work **2.1, 2.12, 2.13, 2.15, 2.17, 2.22, 2.26, and 2.27** in that order (about 70–85 minutes before discussion). This eight-problem route covers span, field-sensitive independence, basis construction and extension, the linear-dependence lemma, coordinate calculation and proof, and a dimension-formula transfer.

Additional practice: 2.2–2.11, 2.14, 2.16, 2.18–2.21, 2.23–2.25, 2.28–2.30. **Bridge:** none required. **Extension:** 2.31.

Span

Exercise 2.1 (computational) Is $(1,2,3)$ in $\operatorname{span}\big((1,1,0),(0,1,1)\big)$ in $\mathbb{R}^3$?

Answer. No: the system is inconsistent, so $(1,2,3) \notin \operatorname{span}\big((1,1,0),(0,1,1)\big)$.

Exercise 2.2 (computational) Is $(2,3,1)$ in $\operatorname{span}\big((1,1,0),(0,1,1)\big)$ in $\mathbb{R}^3$? If so, write it as a combination.

Answer. Yes: $(2,3,1) = 2(1,1,0) + (0,1,1)$.

Exercise 2.3 (computational) For which value of t is $(t,1,2)$ in $\operatorname{span}\big((1,0,1),(1,1,1)\big)$ in $\mathbb{R}^3$?

Answer. $t = 2$.

Exercise 2.4 (computational) In $\mathcal{P}_2(\mathbb{R})$, is $2+5t+3t^2$ in $\operatorname{span}(1+t,\ t+t^2)$?

Answer. Yes: $2+5t+3t^2 = 2(1+t)+3(t+t^2)$.

Exercise 2.5 (proof) Show that $\operatorname{span}(v_1,\dots,v_m) = \operatorname{span}(v_1,\dots,v_m,w)$ if and only if $w \in \operatorname{span}(v_1,\dots,v_m)$.

Linear independence

Exercise 2.6 (computational) Find the value of t for which $(3,1,4),(2,-3,5),(5,9,t)$ is linearly dependent in $\mathbb{R}^3$.

Answer. $t = 2$.

Exercise 2.7 (computational) Is $(1,3),(2,5)$ linearly independent in $\mathbb{R}^2$?

Answer. Yes.

Exercise 2.8 (computational) Is $(1,0,1),(2,1,0),(3,1,2)$ linearly independent in $\mathbb{R}^3$?

Answer. Yes (the determinant is 1).

Exercise 2.9 (computational) Is $1 + t,\ 1 - t,\ t^2$ linearly independent in $\mathcal{P}_2(\mathbb{R})$?

Answer. Yes.

Exercise 2.10 (computational) Is $(1, i), (i, -1)$ linearly independent in $\mathbb{C}^2$?

Answer. No: it is linearly dependent.

Exercise 2.11 (proof) Suppose v_1, v_2, v_3, v_4 is linearly independent in V. Prove that

$$v_1 - v_2, \quad v_2 - v_3, \quad v_3 - v_4, \quad v_4$$

is also linearly independent.

Hint. Set a general combination equal to 0 and gather the coefficient of each v_i; independence of the original list gives a system you can solve one variable at a time.

Exercise 2.12 ($\star$) View $\mathbb{C}$ as a vector space. Show that $1 + i,\ 1 - i$ is linearly independent over $\mathbb{R}$ but linearly dependent over $\mathbb{C}$.

Hint. The scalar field is doing the work. Over $\mathbb{R}$, separate real and imaginary parts; over $\mathbb{C}$, ask what multiplying $1 + i$ by i gives.

Bases and dimension

Exercise 2.13 (computational) Find a basis and the dimension of $U = \{x \in \mathbb{R}^4 : x_1 - x_2 + 2x_3 = 0,\ x_2 - x_3 + x_4 = 0\}$.

Answer. Basis $(1, 1, 0, -1),\ (-2, 0, 1, 1)$; $\dim U = 2$.

Exercise 2.14 (computational) Give a basis and the dimension of (a) the symmetric 2×2 real matrices, and (b) the trace-zero 2×2 real matrices.

Answer. (a) dimension 3; (b) dimension 3.

Exercise 2.15 (computational) Extend the independent list $(1, 2, 1, 0), (0, 1, 1, 1)$ to a basis of $\mathbb{R}^4$.

Answer. E.g. $(1, 2, 1, 0),\ (0, 1, 1, 1),\ (1, 0, 0, 0),\ (0, 0, 1, 0)$ (other choices work).

Exercise 2.16 (computational) Find a basis and the dimension of $U = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$.

Answer. Basis $(-1, 1, 0),\ (-1, 0, 1)$; $\dim U = 2$.

Exercise 2.17 (computational) Find a basis and the dimension of $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 0\}$.

Answer. Basis $t,\ t^2,\ t^3$; dimension 3.

Exercise 2.18 (computational) Find a basis and the dimension of the space of diagonal 3×3 real matrices.

Answer. Basis $\operatorname{diag}(1, 0, 0),\ \operatorname{diag}(0, 1, 0),\ \operatorname{diag}(0, 0, 1)$; dimension 3.

Exercise 2.19 (computational) Use the linear-dependence lemma to remove a redundant vector, then find the dimension of

$$\operatorname{span}\big((1, 2, 3, 4), (2, 4, 6, 8), (1, 0, 1, 0), (0, 1, 0, 1)\big) \subseteq \mathbb{R}^4.$$

Answer. 3.

Exercise 2.20 (proof) Let V be finite-dimensional and $U \subseteq V$ a subspace with $\dim U = \dim V$. Prove that $U = V$.

Exercise 2.21 (proof) Let u_1, u_2, u_3 be a basis of a vector space V. Prove that $u_1,\ u_1 + u_2,\ u_1 + u_2 + u_3$ is also a basis of V.

Coordinates

Exercise 2.22 (computational) Find the coordinate vector of $v = (1, 2, 3)$ in the basis $\mathcal{B} = \big((1, 1, 0), (0, 1, 1), (0, 0, 1)\big)$ of $\mathbb{R}^3$.

Answer. $[v]_{\mathcal{B}} = (1, 1, 2)$.

Exercise 2.23 (computational) Find the coordinate vector of $v = (5, 1)$ in the basis $\mathcal{B} = \big((1, 1), (1, -1)\big)$ of $\mathbb{R}^2$.

Answer. $[v]_{\mathcal{B}} = (3, 2)$.

Exercise 2.24 (computational) Find the coordinate vector of $p = t^2$ in the basis $C = \big(1,\ t - 1,\ (t - 1)^2\big)$ of $\mathcal{P}_2(\mathbb{R})$.

Answer. $[p]_C = (1, 2, 1)$.

Exercise 2.25 (computational) Find the coordinate vector of $v = (2, 3, 4)$ in the basis $\mathcal{B} = \big((1, 0, 0), (1, 1, 0), (1, 1, 1)\big)$ of $\mathbb{R}^3$.

Answer. $[v]_{\mathcal{B}} = (-1, -1, 4)$.

Exercise 2.26 (proof) Let $\mathcal{B}$ be a basis of an n-dimensional space V. Prove that vectors $v_1, \ldots, v_k \in V$ are linearly independent if and only if their coordinate vectors $[v_1]_{\mathcal{B}}, \ldots, [v_k]_{\mathcal{B}}$ are linearly independent in $\mathbb{F}^n$.

Sums and dimension formula

Exercise 2.27 (computational) In $\mathbb{R}^4$ let

$$U = \operatorname{span}\big((1, 1, 1, 1), (1, 0, 1, 0)\big), \qquad W = \operatorname{span}\big((0, 1, 0, 1), (1, 1, 0, 0)\big).$$

Find $\dim(U + W)$ and $\dim(U \cap W)$, and exhibit a basis of $U \cap W$.

Answer. $\dim(U + W) = 3$, $\dim(U \cap W) = 1$, $U \cap W = \operatorname{span}\big((0, 1, 0, 1)\big)$.

Exercise 2.28 (computational)
In $\mathbb{R}^3$ let $U = \operatorname{span}\big((1, 0, 0), (0, 1, 0)\big)$ and $W = \operatorname{span}\big((0, 1, 0), (0, 0, 1)\big)$. Find $\dim(U + W)$ and $\dim(U \cap W)$, and give a basis of $U \cap W$.

Answer. $\dim(U + W) = 3$, $\dim(U \cap W) = 1$, $U \cap W = \operatorname{span}\big((0, 1, 0)\big)$.

Exercise 2.29 (computational) In $\mathbb{R}^4$ let

$$U = \operatorname{span}\big((1, 0, 0, 0), (0, 1, 0, 0)\big), \qquad W = \operatorname{span}\big((0, 0, 1, 0), (0, 0, 0, 1)\big).$$

Find $\dim(U + W)$ and $\dim(U \cap W)$. Is the sum direct?

Answer. $\dim(U + W) = 4$, $\dim(U \cap W) = 0$; the sum is direct.

Exercise 2.30 (computational) In $\mathbb{R}^4$ let

$$U = \operatorname{span}\big((1,1,0,0),(0,0,1,1)\big), \qquad W = \operatorname{span}\big((1,0,0,1),(0,1,1,0)\big).$$

Find $\dim(U+W)$ and $\dim(U \cap W)$, and exhibit a basis of $U \cap W$.

Answer. $\dim(U+W) = 3$, $\dim(U \cap W) = 1$, $U \cap W = \operatorname{span}\big((1,1,1,1)\big)$.

Exercise 2.31 ($\star$) Let V be finite-dimensional with $\dim V = n$, and let U, W be subspaces with $\dim U + \dim W > n$. Prove that $U \cap W \neq \{0\}$.

Hint. Apply the dimension formula to $U+W$, and use that $U+W$ is still a subspace of the n-dimensional V.